MIL-G-14851(MU)
28 April 1971
SUPERSEDING
FA-PD-MI-2565, Rev 0
27 September 1962

MILITARY SPECIFICATION
GENERATOR, GAS PRESSURE, PROPELLANT ACTUATED, M17
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Generator, Gas Pressure, Propellant Actauted, M17, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in the recovery system of training bombs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS
Military
MIL-A-2550 -Ammunition and Special Weapons, General Specification For

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-413 - Visual Inspection Guide for Rubber O-Rings

DRAWINGS
Munitions Command
D10522492 - Generator, Gas Pressure, Propellant Actuated, M17, Metal Parts Assembly

FSC 1377

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Drawings (Cont'd)

C8864324	- Box, Packing, Ammunition for Generator, Gas Pressure, Propellant Actuated, M17 in Fiber Container M510
C8864325	- Container, Ammunition, Fiber M510 for Generator, Gas Pressure, Propellant Actauted, M17
IEL-10522492	- Inspection Equipment List for Generator, Gas Pressure, Propellant Actauted, M17, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document froms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Physical properties.- Physical properties required by the drawings shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.3 Hydrostatic pressure.- The generator chamber body and cap shall withstand an internal hydrostatic pressure of 9000 to 9500 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.2).

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3.4 Shear pin strength.- After application of the protective finish, the pin shall shear when a load of 41 plus or minus 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.5 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of of the component regardless of contour configuration.

3.6 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.7 Workmanship.- The requirements of workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other similar defects.

3.7.2 Burr.- No part shall have a burr which might interfere with the assembly of function of the item, or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- No part or assembly shall contain chips. dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

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4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5 metal parts assemblies, 10 sets of unassembled components, 50 shear pins from the same lot of shear pins tested by the contractor. The sample shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.1.2 Size of lot .- For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 generators, maximum, exclusive of test samples, provided the homogeneity of manufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II

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4.3.2 Examination (Cont'd)

and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table III

Total Allowable Defects per Sample Size

Sample Size for Major Defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Size for Minor Defects	**110**	**150**	**225**	**300**
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

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4.3.2.1 Cap, Initiator, drawing F10522493, covering a detail of drawing F10522492.

	Defect	Method of Inspection
CRITICAL		
	None defined	
101.	Pitch diameter of external thread, min.	Gage
102.	Major diameter of external thread, min.	Gage
103.	Pitch diameter small internal thread, max.	Gage
104.	Minor diameter of small internal thread, max.	Gage
105.	Pitch diameter of large internal thread, max.	Gage
106.	Minor diameter of large internal thread, max.	Gage
107.	Concentricity pitch diameter of large internal thread with long cavity	Gage
108.	Concentricity of firing pin cavity with long cavity	Gage
109.	Concentricity small external thread with long cavity	Gage
110.	Concentricity of body shoulder with pitch diameter of large internal thread	Gage
111.	Perpendicularity of internal boss face with pitch diameter large internal thread	Gage
112.	Symmetry of shear pin hole with long cavity	Gage
113.	Diameter of long cavity	Gage
114.	Diameter of firing pin cavity	Gage
115.	Diameter of mouth at small threaded cavity	Gage
116.	Diameter of counterbore at mouth of small threaded cavity	Gage
117.	Diameter of shear pin hole, max.	Gage
118.	Depth to cartridge head recess	Gage
119.	Depth of set screw hole to angle of chamfer	Gage
120.	Depth of counterbore at mouth of small threaded cavity	Gage
121.	Length of effective large internal thread, min.	Gage
122.	Length of effective small internal thread, min.	Gage
123.	Length of imperfect thread, max.	Gage
124.	Height of boss	Gage
125.	Thread damaged or not full	Gage
126.	Metal defective	Visual
127.	Protective finish inadequate or defective	Visual
128.	Surface finish of long cavity and mouth of threaded cavity	Visual
129.	Burr or sharp corner	Visual
130.	Foreign matter	Visual

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4.3.2.1 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of small internal thread, min.	Gage
203.	Pitch diameter of large internal thread, min.	Gage
204.	Pitch diameter shear pin set screw thread	Gage
205.	Minor diameter shear pin set screw thread, max.	Gage
206.	Pitch diameter set screw thread	Gage
207.	Minor diameter set screw thread, max.	Gage
208.	Diameter of body shoulder	Gage
209.	Diameter of body	Gage
210.	Diameter of undercut	Gage
211.	Diameter of shear pin flange	Gage
212.	Diameter of threaded cavity at thread relief	Gage
213.	Diameter of shear pin hole, min.	Gage
214.	Diameter of boss at datum location	Gage
215.	Depth to firing pin cavity base	Gage
216.	Depth to base of threaded cavity including chamfer	Gage
217.	Length of neck	Gage
218.	Length of effective set screw thread, min.	Gage
219.	Width of mounting flat	Gage
220.	Location of set screw hole	Gage
221.	Location of shear pin flange	Gage
222.	Location of set screw hole	Gage
223.	Angular location from set screw holes	Gage
224.	Angle at datum location	Gage
225.	Diameter of hole between firing pin cavity and threaded cavity	Gage
226.	Depth of mounting flat	Gage
227.	Total length	Gage
228.	Length of shoulder on body	Gage
229.	Width across neck	Gage
230.	Width of undercut	Gage
231.	Width across flat	Gage
232.	Location of body	Gage
233.	Location of undercut	Gage
234.	Surface finish, improper except as otherwise classified	Visual

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4.3.2.2 Pin, Firing, drawing C10522494 covering a detail of drawing F10522492.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of "O" ring groove with body	Gage
102. Concentricity of striker with body	Gage
103. Concentricity of small body with body	Gage
104. Perpendicularity of face of striker with body	Gage
105. Symmetry of shear pin hole with body	Gage
106. Diameter of body	Gage
107. Diameter of shear pin hole, max.	Gage
108. Depth of shear pin hole, max.	Gage
109. Length of striker, min.	Gage
110. Profile of groove	Gage
111. Metal defective	Visual
112. Protective finish, inadequate or defective	Visual
113. Surface finish of body, improper	Visual
114. Burr or sharp corner	Visual
115. Foreign matter	Visual
MINOR	
201. Diameter of small body	Gage
202. Diameter of shear pin hole, min.	Gage
203. Diameter of "O" ring groove	Gage
204. Diameter of striker	Gage
205. Depth of shear pin hole, min.	Gage
206. Depth of slot	Gage
207. Length of striker, max.	Gage
208. Location of shear pin hole	Gage
209. Location of slot	Gage
210. Total length	Gage
211. Length of small body	Gage
212. Width of slot	Gage
213. Width of "O" ring groove	Gage
214. Location of "O" ring groove	Gage
215. Radius or chamfer, missing or incorrect	Visual
216. Surface finish, improper except as otherwise classified	Visual

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4.3.2.3 Pin, Shear, drawing B10522495, covering a detail of drawing F10522492.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish, inadequate or defective	Visual
105. Surface finish, improper	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Profile of end	Gage

4.3.2.4 Chamber, body, drawing F10522496, covering a detail of drawing F10522492.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of large thread, min.	Gage
102. Major diameter of large thread, min.	Gage
103. Pitch diameter of small thread, min.	Gage
104. Major diameter of small thread, min.	Gage
105. Concentricity of counterbore with large inside diameter	Gage
106. Concentricity of pitch diameter of large thread with large inside diameter	Gage
107. Concentricity of thread pilot with pitch diameter of small thread	Gage
108. Concentricity of step in front of thread pilot with pitch diameter of small thread	Gage
109. Diameter of large cavity	Gage
110. Thickness of wall of body	Gage
111. Length of thread	Gage
112. Thickness of wall of body	Gage
113. Permissible variation in sidewalls of groove	Gage

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	Defect	Method of Inspection
114.	Thread damaged	Visual
115.	Metal defective	Visual
116.	Protective finish inadequate or defective	Visual
117.	Burr or sharp corner	Visual
118.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of alrge thread, max.	Gage
202.	Pitch diameter of small thread, max.	Gage
203.	Diameter of small cavity	Gage
204.	Diameter of body	Gage
205.	Diameter of step in front of small thread pilot	Gage
206.	Diameter of counterbore	Gage
207.	Diameter of pellet	Gage
208.	Diameter of groove in step	Gage
209.	Diameter over top of pellet	Gage
210.	Depth of counterbore	Gage
211.	Depth to cavity taper	Gage
212.	Length of small thread chamfer	Gage
213.	Length of step in front of small thread pilot	Gage
214.	Length of large thread pilot	Gage
215.	Length of body	Gage
216.	Total length	Gage
217.	Width of groove in step in front of small thread pilot	Gage
218.	Location of pellet	Gage
219.	Angular location of pellet	Gage
220.	Location of groove in step in front of small thread pilot	Gage
221.	Location of small thread	Gage
222.	Length of chamfer	Gage
223.	Location of large thread	Gage
224.	Pellet radius incorrect	Visual
225.	Pellet missing, loose or improperly inserted	Visual
226.	Radius or chamfer missing or incorrect	Visual
227.	Surface finish, improper	Visual

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4.3.2.5 Assembly, drawing D10522492

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Improper assembly	Visual
103. "O" ring not lubricated	Visual
104. Marking missing, incorrect or illegible	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Shipping cap missing	Visual
202. Protective finish inadequate or defective	Visual

4.3.2.6 Fiber Container, drawing C8864325.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.7 Box, Packing, drawing C8864324

CRITICAL	
None defined	
MAJOR	
None defined	

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units, incorrect	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested for compliance with the requirements specified on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 Hydrostatic pressure.- Each generator chamber body and cap subassembly shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with the requirements of 3.3 shall be cause for rejection of the component or subassembly.

4.3.3.3 Shear pin strength.-Fifty pins from each lot of shear pins shall be tested by a government approved laboratory for shear strength. Failure of any pin to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.4 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standard listed on the drawing.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-F10522492.

4.4 Test methods and procedures.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness testing machine shall be used to determine compliance with the hardness requirements specified on the detail drawing.

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4.4.2 Hydrostatic pressure.- Prior to the application of protective finish, the chamber and cap subassembly shall be mounted in the fixture using the devices for properly sealing all openings and subjected to the hydrostatic pressure to determine compliance with 3.3.

4.4.3 Shear pin strength.- The test shall be performed after application of protective finish. The pin shall be inserted in the fixture and tested in conjunction with a Tensile Testing Machine. The load shall be applied at the rate of 0.125 inches per minute.

4.4.4 Protective finish.- Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.4.5 Residual magnetism.- The generator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the generator 6 inches from, and at the north-south horizontal position with the compass set in an east-west heading. The generator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position and moved slowly past the compass.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with drawings C8864324 and C8864325.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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